XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE312 Week 8
Lab Practice
Implementation of the Bisection Method in C Programming Language

Write a C program to find the roots of the following nonlinear equation:

$$f(x) = -3\cos^2 x + 3\sin x, x \in [-6, 2].$$

Use the bisection method.

Summarize the results at every iteration by printing the values of a_i, b_i, c_i and $|c_i - c_{i-1}|$ as the iteration progresses.

The stopping condition is until $|c_i - c_{i-1}| < \varepsilon$ where, where ε = 0.000001.

.

// Beginning of the C Source code

```
1   // bisection method
2   #include <stdio.h>
3   #include <math.h>
4
5   float nonlinearFunction(float x)
6   {
7       float y;
8       // Write your code here…
9       y = -3.0*pow(cos(x),2) + 3.0*sin(x);
10
11      return y;
12  }
```

Line 5-11: Implement the nonlinear function

$$f(x) = -3\cos^2 x + 3\sin x.$$

Function main()

```
14  int main()
15  {
16      float a = -6.0; // initialize endpoint on the left
17      float b = 2.0; // initialize endpoint of the right
18      float c = 0.0; // the midpoint between a and b
19      float old_c = 0.0; // store the previous iteration of c
20      float f_a, f_c; // variables to store the
21              //function values evaluated at a and c
22      float epsilon = 0.000001; // stop the program if
23                                // fabs(c-old_c) is less than 0.000001
24      float error = 100; // store the error = fabs(c-old_c)
```

```
26      do {
27          old_c = c; // store the value c in old_c
28          c = (a+b)/2.0; // calculate the midpoint c
29
30          printf("\na=%f, b=%f, c=%f", a, b, c);
31
32          f_a = nonlinearFunction(a);
33          f_c = nonlinearFunction(c);
34
```

Line 27: Store the previous value of cin old_c.

Line 28: Calculate the midpoint c= (a+b)/2.

Line 30: Print out the values of a, band c.

Line 32 and 33: Evaluate the nonlinear function at x= aand x= cand store them in variables f_a and f_c, respectively.

```
35          if(f_a*f_c<0)
36          {
37              b = c;
38          } else if(f_a*f_c>0) {
39              a = c;
40          }
```

Line 35-39: Depending on the value of f_a*f_c, set the values of endpoints a and b according to the following conditions:

If $f_a*f_c < 0$ then

Update $b_{i+1} = c_i$ and $a_{i+1} = a_i$

else if $f_a*f_c > 0$ then

Update $a_{i+1} = c_i$ and $b_{i+1} = b_i$

```
42          error = fabs(c-old_c);
43          printf("\nError=%f", error);
44
45      } while (error>epsilon);
46
47      printf("\nThe root of the nonlinear equation is x = %f.", c);
48
49      return 0;
50  }
```

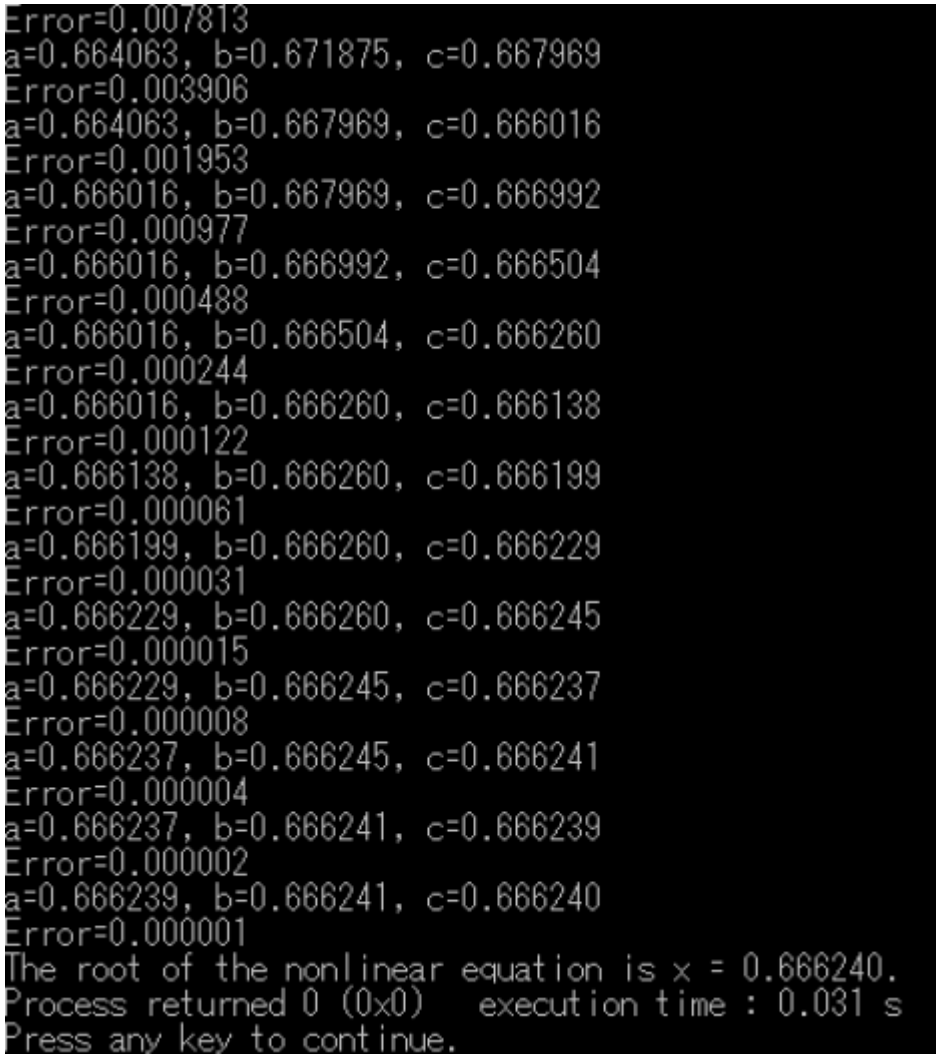
Line 42: Calculate the error which is the absolute difference between the current value of c and the previous iteration value old_c.

Line 43: Print out the error.

Line 45: If the difference is less than epsilon, stop the iteration and output the final solution. Otherwise, return to the beginning of the do-while loop and repeat the iteration.

Line 47: Print the final result of the root of the nonlinear equation.

Output


Error=0.007813
a=0.664063, b=0.671875, c=0.667969
Error=0.003906
a=0.664063, b=0.667969, c=0.666016
Error=0.001953
a=0.666016, b=0.667969, c=0.666992
Error=0.000977
a=0.666016, b=0.666992, c=0.666504
Error=0.000488
a=0.666016, b=0.666504, c=0.666260
Error=0.000244
a=0.666016, b=0.666260, c=0.666138
Error=0.000122
a=0.666138, b=0.666260, c=0.666199
Error=0.000061
a=0.666199, b=0.666260, c=0.666229
Error=0.000031
a=0.666229, b=0.666260, c=0.666245
Error=0.000015
a=0.666229, b=0.666245, c=0.666237
Error=0.000008
a=0.666237, b=0.666245, c=0.666241
Error=0.000004
a=0.666237, b=0.666241, c=0.666239
Error=0.000002
a=0.666239, b=0.666241, c=0.666240
Error=0.000001
The root of the nonlinear equation is x = 0.666240.
Process returned 0 (0x0) execution time : 0.031 s
Press any key to continue.